

Jazan University, Department of Chemistry
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Sarvat Zafar, Ph.D

Current Position

Worked as Assistant Professor of Chemistry in Jazan University, Jazan, Kingdom of Saudi Arabia.

Professional Experience (10 years)

- Assistant Professor of **Chemistry**, Jazan University **2010-2020**
  Jazan, Kingdom of Saudi Arabia.
- Online Teaching Experience (**Blackboard Tutorial App**) **2017-2019**
- **Tenth Scientific Forum Grant**, Jazan University, Jazan KSA **2019-2020**
  Developed and characterized (FT-IR, UV-vis, SEM, and TEM) gold and silver nanoparticles through biological green technique and products was also tested towards human pathogenic bacterial strains and their potential cytotoxicity in human cancer cell lines (MCF-7).
- University Grants Commission (**UGC**) Fellowship **2007-2008**
  Govt. of India, New Delhi, India
  Developed and characterized (FT-IR,^{1}H-NMR,^{13}C-NMR, anticorrosive and antimicrobial properties) eco-friendly and sustainable products based waterborne epoxy paints and coatings.
- Senior Research Fellow (**SRF**), Jamia Millia Islamia **2008-2009**
  Govt. of India, New Delhi, India
  Developed and studied spectroscopic and morphological properties of Resorcinol-formaldehyde based waterborne poly(1-naphthylamine/polyvinyl) based corrosion resistant composite coatings.
- Senior Research Fellow (**SRF-Ext**), Jamia Millia Islamia **2009-2010**
  Govt. of India, New Delhi
  Synthesized Oleic acid based waterborne interpenetrating polymers and their characterization (FT-IR, ^{1}H-NMR, ^{13}C-NMR, SEM, anticorrosive and antimicrobial properties).

Education

Doctor of Philosophy (Ph.D), Polymer Chemistry **2009**

Department of Chemistry, Jamia Millia Islamia, Central University, New Delhi, India, *Ph.D topic* ***"Development of Sustainable Resource Based Waterborne Coating Material"***

Postgraduate Diploma in Chemo-informatics **2006**

Hamdard University, New Delhi, India

Master of Science (M.Sc.), Chemistry (Organic Chemistry) **2002**

Department of Chemistry, Aligarh Muslim University, Aligarh, India

Bachelor of Science (B.Sc. Hons.), Chemistry **2000**

Department of Chemistry, Aligarh Muslim University, Aligarh, India

Research Experience

i. Synthesis of polymers using **Green Approach** and their spectroscopic and morphological studies.
ii. Strategic Approaches for dispersing nanoparticles in **Green** synthesized polymeric matrix.
iii. Synthesis of **Metallic Nanoparticles** (Ag, Au, ferrites, etc.) from various plant sources and their antimicrobial and cytotoxic activities towards numerous cancer cell lines.
iv. Synthesis of **Graphene** Nanoparticles and their use in water desalination technologies.

Teaching Experience

Courses Taught (since 2010)

- Aliphatic Chemistry (1st year undergraduate students)
- Aromatic Chemistry (3rd year undergraduate students)
- Heterocyclic Chemistry (3rd year undergraduate students)
- Polymer Chemistry (4th year undergraduate students)
- Applied Organic Chemistry (4th year undergraduate students)
- Stereochemistry (4th year undergraduate students)

- Organic Reaction Mechanism (3rd year undergraduate students)
- Laboratory instructions (4th year undergraduate students)

Administrative Work

- Member of NCAAA for ***standard 3***-Teaching and learning (2018-2020)
- Member of NCAAA for ***standard 10***-Research (2016-2018)

Awards and Fellowships

- **UGC (University Grants Commission)** Fellowship, Govt. of India, 2007-2008.
- **CSIR (Council of Scientific and Industrial Research)** Senior Research Fellowship, Govt. of India, 2008-2009.
- **CSIR (Council of Scientific and Industrial Research)** Senior Research Fellowship-Extended, Govt. of India, 2009-2010.
- Poster presentation award in International conference Polychar-2009.

Analytical Techniques Known

- XRD, SEM, TEM, TGA, DTA, DSC
- Spectroscopic Techniques (NMR, FT-IR, UV-Visible, Mass)
- Chromatography Techniques (GLC, GSC, HPLC, TLC)
- Conductometry, Potentiometry, pH-metery

Computer Proficiency

Good computer proficiency in MS-DOS, Window operating system, Microsoft office, MS Word, Excel, PowerPoint, Adobe Photoshop, illustrator, End note, Languages - Basic, Database and Internet, Emailing.

List of Publications

1. Waterborne melamine formaldehyde cured Epoxy-acrylate corrosion resistant coatings.
   Sarvat Zafar, Uffana Riaz and Sharif Ahmad
   Journal of Applied Polymer Science, 107(1), 215-222, 2008

2. Development of novel water borne poly (1-naphthylamine)/ poly(vinyl alcohol)-resorcinol formaldehyde cured corrosion resistant composite coatings.
   Sarvat Zafar, Ufana Riaz, S.M.Ashraf, and Sharif Ahmad
   Progress in Organic coatings 62(1), 32-39, 2008

3. Synthesis, characterization and anticorrosive coating properties of waterborne interpenetrating polymer network based on epoxy-acrylic-oleic acid with butylated melamine formaldehyde.
   Sarvat Zafar, Ufana Riaz, and Sharif Ahmad
   Journal of Applied Polymer Science, 113, 827-838, 2009

4. Various effects of formalin amongst the health workers - A questionnaire-based study.
   Sarvat Zafar and Aiman Zafar
   World Journal of Pharmacy and Pharmaceutical Sciences 8, 2018

5. Waterborne Epoxy based Coating Materials
   Sarvat Zafar
   American Scientific Research Journal for Engineering, Technology, and Sciences, 50, 2018

6. Biosynthesis and Characterization of Silver Nanoparticles Using *Phoenix dactylifera* Fruits Extract and their *In Vitro* Antimicrobial and Cytotoxic Effects
   Sarvat Zafar and Aiman Zafar
   The Open Biotechnology Journal 13, 2019

Conferences Attended

1. **Recent Advanced in Chemistry 2008**, Jamia Millia Islamia, New Delhi India
2. **Natural Science Fest 2008,** Jamia Millia Islamia, New Delhi, India.
3. **Interdisciplinary Science Conference –2007 on Recent Trends of Research in Biological Science ISC –2007,** Jamia Millia Islamia New Delhi, India.
4. **Applications of Conducting Polymers and Nanomaterials in science &Technology 2006,** Jamia Millia Islamia, New Delhi India.

Workshop/Symposium

1. ***"How to become a productive writer"*** organized by Faculty of Science, UTM, Johar Bahru, Malaysia, 22 August 2017,
2. ***"Recent advances in chemistry"*** organized by Department of chemistry, Jamia Millia Islamia, New Delhi, India, January 19, 2009
3. ***"Recent trends in research in biological sciences"*** organized by Centre for Interdisciplinary research in basic sciences, Jamia Millia Islamia, New Delhi, India, December 7, 2007
4. ***"Synthesis and characterization of Nanomaterials"*** organized by Indian Society of Analytical Scientists and Department of chemistry, Jamia Hamdard, New Delhi, India, November 17, 2007
5. ***"Chromatographic techniques"*** organized by Indian Society of Analytical Scientists and Department of chemistry, Jamia Millia Islamia, New Delhi, India September 10, 2007

6. ***"Applications of NMR to materials and medicine"*** organized by Department of Chemistry, IIT, New Delhi, India, November 21, 2006

References

1. **Dr. Fakhra Jameel** (Assistant Professor)
College of Science Department of Chemistry Jazan University, KSA
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2. **Dr. Manal M.Rikabi** (Associate Professor)
College of Science Department of Chemistry Jazan University, KSA
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3. **Dr. Miad Ali Siddiq** (Assistant Professor)
College of Science Department of Chemistry Jazan University, KSA
Mob: 00966- 546456328
Email: msiddiq@jazanu.edu.sa

4. **Dr. Rabab M. Nasser** (Assistant Professor)
College of Science Department of Chemistry Jazan University, KSA
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Email: rabab nasser@yahoo.com

Cover Letter

Dear Sir/Madam,

I would like to apply for the Position of **Senior Lecturer** in your esteemed university. I have done my Ph.D (Applied Organic Chemistry) from Jamia Millia Islamia, New Delhi, India. The title of my thesis was ***"Development of sustainable resource based waterborne coating material".*** I have publications in ISI reputed journals as well as other international journals in ***eco-friendly organic polymers and metal nanoparticles.*** I have also participated in National and International Conferences. I was awarded Junior Research Fellowship and Senior Research Fellowships from UGC-CSIR between **2007 and 2010**. I worked as Assistant Professor of Chemistry in **Jazan Univeristy, Jazan KSA (2010-2020)** having **ten years** teaching experience dealing with undergraduate students.

In consonance with your job requirements, my profile of skill and achievements are included in my resume, which is enclosed herewith for yours kind and considerate perusal.

I am sure that my skill and experience are suitable with your job requirements.

Thanking you and with kind regards,

Yours faithfully,

Dr. SARVAT ZAFAR
Department of Chemistry
Jazan University,
Jazan, KSA
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Teaching Philosophy

I believe in teaching philosophy of not just teaching but inspiring the students for learning through insightful behavior on one side and to evolve the curiosity in the students towards the discipline. On the other hand, my firm conviction is that the motivation and encouraging contacts leads to the state of affairs, where bright students may excel and support to the struggling ones as well.

Conveying thrill and excitement in the taught in enthusiastic manner will create intrinsic interest of the students in the discipline of chemistry and not only just showing and illustrating by words and the signs. I communicate the integrations of chemistry and life. For an inspiring faculty genuine interest and enthusiasm besides the command is necessary only then I can inculcate in the student's relevance of the subject of chemistry having innovative approach in experiments and procedure.

High expectation is my goal. From day one I make it explicit to my students that I would like them to strive for excellence through reciprocity and critical valence. I always like to alleviate the students from bottom to top by boosting the level of self-confidence.

I prefer a relaxed atmosphere in the classroom so that even the back benches may feel comfortable with high academics at far. I also maintain the student-teacher interaction in classroom and outside providing them advice and assistance in any way that I can believe in the principle of turning geese into swans through the harm one of encouragement.

Teaching Interests

I have taught many courses of Organic Chemistry since last 10 years as an Assistant Professor. The courses comprises: Aliphatic Chemistry (1st year undergraduate students), Aromatic Chemistry (3rd year undergraduate students), Heterocyclic Chemistry (3rd year undergraduate students), Polymer Chemistry (4th year undergraduate students), Applied Organic Chemistry (4th year undergraduate students), Stereochemistry (4th year undergraduate students), Organic Reaction Mechanism (3rd year undergraduate students), Laboratory instructions (4th year undergraduate students), Conducted research with undergraduates on applied organic chemistry.

I would like to teach the following courses at both undergraduate and graduate levels: Organic Chemistry, Polymer Chemistry, Applied Organic Chemistry. I could also teach **"Nanomaterials"** or other courses that match my educational and research interest.

RESEARCH STATEMENT

Research Approach

Numerous strategic approaches have been employed for innovative research on the development of alternative green and renewable techniques. ***Green chemistry*** is advantageous not only because of eco-friendlier approaches but it can also reduce or eliminate the uses and generation of hazardous substances in the design, processing, and application of chemical products. Added to this, considerable efforts have been made by the commercial industries and research scientists to avoid the usage of synthetic chemical-based procedures. Hence, based on my research interest or

background, I would like to propose biological or green methods for the development of sustainable products without using noxious chemical ingredients.

Future Prospects

I would like to propose cost-effective, clean, fast and bio-green process for the development of sustainable green products in my research work, for instance, (a) Polymer based Nanocomposites, (b) Metallic nanoparticles, (c) Antimicrobial and cytotoxic activities, (d) water desalination studies.

My Future prospects include:

i. Synthesis of polymers using **Green Approach** and their spectroscopic and morphological studies.
ii. Strategic Approaches for dispersing nanoparticles in **Green** synthesized polymeric matrix.
iii. Synthesis of **Metallic Nanoparticles** (Ag, Au, ferrites, etc.) from various plant sources and their antimicrobial and cytotoxic activities towards numerous cancer cell lines.
iv. Synthesis of **Graphene** Nanoparticles and their use in water desalination technologies.

Conclusion

My research activities include development of sustainable resource-based waterborne green polymers using deionized water as co-solvent and studied their anticorrosive and antimicrobial activities. Spectroscopic techniques (^{1}H-NMR, ^{13}C-NMR, FT-IR, TEM and SEM) are also studied for structure elucidation and surface morphology. I am also interested in developing metallic nanoparticles from numerous plant sources and investigated their cytotoxic and antimicrobial activities. Strategic approaches for production of graphene-based nanoparticles and their application in water desalination technologies are also undergoing research activities.